

Orthogonal Dual-Tree Topologies for Fault-Tolerant Distributed Aggregation

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Abstract

Tree-based communication topologies are widely used in distributed machine learning and distributed aggregation systems due to their logarithmic communication depth and efficient hierarchical structure. However, conventional tree topologies suffer from poor fault tolerance because failure of an internal node disconnects entire dependent subtrees.

This work proposes a preliminary communication topology based on two complementary spanning trees constructed over the same logical node set. The proposed design attempts to reduce persistent node criticality by assigning complementary hierarchical roles across the two trees. Nodes that are highly critical in one tree are intentionally placed in lower-criticality positions in the second tree.

Unlike conventional redundant tree approaches, this work additionally introduces explicit inter-tree identity connections between corresponding node instances across the two trees. This layered dual-tree structure enables alternate routing paths while preserving hierarchical communication efficiency.

This document presents the preliminary formulation, intuition, structural definitions, and initial theoretical direction of the proposed topology.

1 Introduction

Distributed machine learning systems commonly employ hierarchical communication structures for operations such as:

- gradient aggregation,
- parameter synchronization,
- reduce and all-reduce operations,
- hierarchical broadcast.

Tree-based communication structures are attractive because they provide:

- logarithmic communication depth,
- scalable aggregation,
- low communication overhead.

However, conventional trees suffer from a major limitation:

Failure of an internal node disconnects all dependent descendants.

This creates a significant reliability issue for large-scale distributed systems and distributed machine learning workloads.

This work proposes a preliminary dual-tree topology intended to reduce such failure sensitivity.

2 Layered Dual-Tree Topology

Let:

$$V = \{v_1, v_2, \dots, v_n\}$$

be the logical node set.

Two tree layers are constructed:

$$T_1 = (V_1, E_1)$$

$$T_2 = (V_2, E_2)$$

where:

$$V_1 = \{v_1^{(1)}, v_2^{(1)}, \dots, v_n^{(1)}\}$$

$$V_2 = \{v_1^{(2)}, v_2^{(2)}, \dots, v_n^{(2)}\}$$

represent node instances in Tree 1 and Tree 2 respectively.

2.1 Inter-Tree Identity Connections

For every logical node v_i , an explicit inter-tree connection is introduced between its two layer instances:

$$E_c = \{(v_i^{(1)}, v_i^{(2)}) \mid 1 \leq i \leq n\}$$

These edges represent identity-preserving transitions between the two trees.

The complete communication graph is therefore:

$$G = (V_1 \cup V_2, E_1 \cup E_2 \cup E_c)$$

This formulation converts the topology into a layered redundant communication graph rather than a conventional single-tree hierarchy.

3 Core Design Principle

The central idea of the proposed topology is:

Node criticality should be anti-correlated across the two trees.

Informally:

- Nodes highly critical in T_1 should become minimally critical in T_2 .
- Nodes near the periphery of T_1 should become structurally important in T_2 .

This attempts to avoid persistent bottleneck nodes across the combined communication structure.

3.1 Initial Role-Inversion Intuition

An initial motivating intuition is:

$$L(T_1) \approx I(T_2)$$

$$I(T_1) \approx L(T_2)$$

where:

- $L(T)$ denotes leaf nodes,
- $I(T)$ denotes internal nodes.

Exact inversion may not always be feasible under strict binary tree constraints. Therefore, approximate inversion or criticality inversion becomes the practical design objective.

4 Node Criticality

Define a node criticality function:

$$C(v, T)$$

which measures the structural importance of node v within tree T .
Possible definitions include:

- subtree size,
- betweenness centrality,
- ancestor dependency count,
- distance from root.

The proposed topology seeks to minimize persistent criticality across both trees.
A preliminary objective function is:

$$\sum_{v \in V} C(v, T_1) C(v, T_2)$$

Minimizing this objective discourages nodes from simultaneously becoming highly critical in both trees.

5 Structural Diversity

Fault tolerance depends not only on redundancy, but also on structural diversity between the trees.

The proposed topology therefore attempts to minimize overlap between:

- ancestor relationships,
- articulation structures,
- routing dependencies,
- communication bottlenecks.

The intention is that local failures in one tree become non-local in the second tree.

6 Fault-Tolerance Intuition

Consider failure of an internal node in T_1 .

In a conventional tree:

- all descendants become disconnected.

In the proposed dual-tree structure:

- descendants may transition through inter-tree identity edges,
- reroute through alternate dependency structures in T_2 ,
- and potentially reconnect through unaffected regions of the communication graph.

Thus the combined layered graph may remain connected even after single-node failures.

7 Failure Model

A logical node failure is assumed to remove both layer instances of the node:

$$v_i^{(1)}$$

and

$$v_i^{(2)}$$

simultaneously.

Thus failures represent physical node failures rather than isolated layer failures.

The topology attempts to maintain communication despite such coupled failures by ensuring dependency diversity across the two trees.

8 Preliminary Theoretical Direction

A central research question is whether the combined graph:

$$G = (V_1 \cup V_2, E_1 \cup E_2 \cup E_c)$$

can satisfy:

$$\kappa(G) \geq 2$$

where:

- $\kappa(G)$ denotes graph vertex connectivity.

If true, the topology would remain connected after removal of any single logical node.

A preliminary conjecture is:

If the two trees maintain sufficiently distinct dependency structures and anti-correlated node criticality, then the combined graph may achieve improved fault tolerance compared to conventional tree topologies.

Formal proof and exact construction conditions remain open problems.

9 Illustrative Example

Figure 1 illustrates the conceptual layered dual-tree structure.

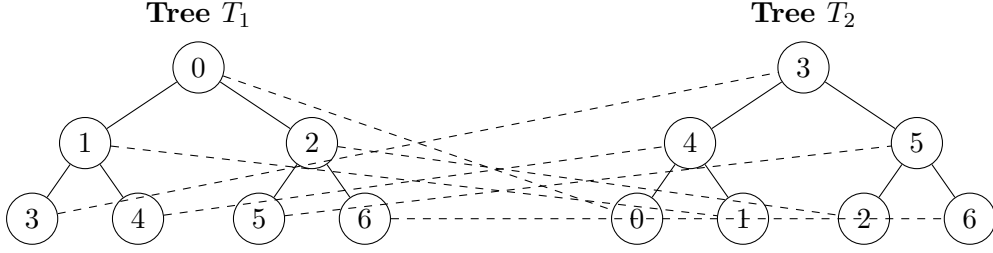


Figure 1: Conceptual layered dual-tree topology with explicit inter-tree identity connections. Dashed lines represent inter-tree transitions between corresponding node instances.

10 Potential Applications

Potential applications include:

- distributed machine learning,
- hierarchical aggregation,
- fault-tolerant overlays,
- distributed consensus,
- resilient communication fabrics.

11 Open Problems

Several important problems remain open:

- optimal dual-tree construction algorithms,
- formal connectivity proofs,
- minimal-overlap hierarchy generation,
- communication latency analysis,
- dynamic rerouting algorithms,
- comparison with ring and fat-tree topologies,
- resilience under multiple simultaneous failures.

12 Conclusion

This work introduced a preliminary layered dual-tree communication topology intended to improve fault tolerance in hierarchical distributed systems.

The key idea is the construction of complementary spanning trees with anti-correlated node criticality and explicit inter-tree identity connections, reducing persistent bottlenecks and improving alternate communication path availability.

The proposed topology remains an early-stage research direction. Future work includes formal graph-theoretic analysis, construction algorithms, simulations, and distributed machine learning evaluation.

References

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- [2] Ian Foster, *Designing and Building Parallel Programs*.
- [3] George Karypis and Vipin Kumar, “Fault-Tolerant Communication Topologies in Distributed Systems.”